Association Rule Mining

What is Data Mining?

It is the exploration and analysis of large quantities of data to discover meaningful rules and patterns. One of the most important techniques for data mining is association rule mining. Other methods are classification, clustering, regression etc.

Association Rule Mining:

- Used to find patterns and correlations in data sets
- Useful for product placement, promotions, stock management
- Also known as Market Basket Analysis

Applications of Association Rule Mining:

- Cross-selling - suggests you to buy additional products that go well with your existing purchase.
- Optimal layout of stores, catalogues, and webpages
- Selecting products for promotions
- Product placement and affinity positioning - placing products that are likely to be purchased together close to one another
- Operations management - helps by identifying items often bought together, so these can be stocked in the same centers, enabling faster deliveries with fewer shipments

Differences From Recommendation Systems

Association rule mining finds patterns in big sets of transaction data. Recommendation systems look at your specific preferences; past behaviour etc. and then suggest products tailored just for you. It often uses association rules and other methods like collaborative filtering.

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Other Considerations

- Sequential Pattern Matching: It focuses on sequences of purchases across multiple transactions. It is useful for understanding customer behaviour over time, such as repeat purchases or evolving needs.
- If A, then B doesn't mean if B, then A.

Key Concepts

- It works on the concept of IF-THEN statements. The "if" part is called the antecedent, and the "then" part is the consequent.
- General Notation: {item i, item j} → {item k, item l}
- Two key problems while developing good rules:
  - Computation: Searching for patterns in large databases is computationally expensive, and there are many potential rules to evaluate.
  - Evaluation: How do we determine whether a rule is "good" or "potentially useful"?
- Practical Solution – only consider items that occur with high frequency in the database (frequent itemsets).
- If the set {item A, item B} is not frequent, then no set containing both item A and item B is frequent. Any subset of a frequent itemset is also a frequent itemset. These are also known as the Apriori Principles.

Support

- Support = $\frac{\text{Frequency of an itemset}}{\text{Total no. of transactions in the data set}}$
- It is the probability that a randomly selected transaction will contain all the items of both the antecedent and consequent sets, i.e. support {A} → {B} = P(A∩B).
- Support of {A} → {B} = support of {B} → {A} since P(A∩B) = P(B∩A).

Confidence

- Confidence = $\frac{\text{No. of transactions with all items in both itemsets}}{\text{No. of transactions with the antecedent itemsets}}$
- It is the conditional probability that the items in the consequent set are present in a randomly selected transaction that includes the antecedent set.
- Confidence of {A} → {B} may not be equal to confidence of {B} → {A}.

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Lift Ratio

- Lift Ratio = $\frac{\text{Confidence}}{\text{Benchmark confidence}}$
- Benchmark Confidence = P(consequent) = confidence assuming antecedent sets and consequent sets are independent.
- A lift > 1 means items appear together more often than by chance, indicating a meaningful association while a lift ≤ 1 suggests weak or no real association.

Importance of Lift Ratio

- Confidence alone can be misleading.
- If the consequent is very common, confidence might be high even if there's no real association.
- Lift solves this by comparing the observed co-occurrence of items to what would be expected if they were independent.

Leverage

- Leverage(A→B) = Support(A→B) − Support(A) × Support(B)
- It measures the difference between the observed frequency of co-occurrence of items and the expected frequency if the items were independent.
- A leverage value of 0 means the items are independent while a positive value indicates that the items co-occur more frequently than expected, showing a potentially meaningful association (and vice-versa).

Pitfalls to Consider

- High support or high confidence doesn't imply causation.
- Low-support rules may be uninteresting from a business perspective.
- Exception: If the consequent is highly valuable and/or the confidence is very high, then you may want to consider the rule.
- Confidence and support thresholds depend on data.
- Large numbers of rules can be generated, including trivial or redundant ones that may be hard to interpret.

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- Statistical significance and domain knowledge are necessary; high confidence or lift alone does not guarantee a useful or actionable insight.

Apriori Algorithm

- A popular method used in association rule mining to find frequent itemsets and derive strong association rules.
- It works by iteratively identifying frequently occurring itemsets and then building larger sets step-by-step.
- Support is used to identify frequent itemsets, whereas confidence filters strong association rules among those itemsets.
- Lower min. support/confidence finds more rules (risking noise); higher min. support/confidence finds fewer but stronger patterns.
- Setting appropriate min-support and min-confidence thresholds balances the discovery of useful patterns with computational efficiency.

Steps in Apriori Algorithm

- Set Minimum Support and Confidence
- Find Frequent 1-Itemsets
- Generate Candidate k-Itemsets
- Prune Candidates
- Count Support of Candidates
- Repeat Steps 3–5
- Generate Association Rules

Other Useful Pointers

- Candidate generation: Generate new candidate itemsets by joining frequent itemsets found in the previous iteration.
- Pruning: Remove candidates that have infrequent subsets based on the Apriori principle, avoiding unnecessary counting in the data. Pruning avoids checking unnecessary itemsets.
- Tuning min. support:
  - Start with a reasonable initial threshold based on domain knowledge

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  - Iterative process, run the algorithm with different threshold values
  - For data sets with diverse item frequencies, multiple minimum supports can be set
- Tuning min. confidence:
  - Choose an initial confidence threshold that reflects acceptable reliability
  - Iterative process; depends on data and domain needs
- Domain expertise often guides initial choices and final tuning decisions.
- Additional metrics such as lift can be used for further filtering.

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